Seeking Commonality While Preserving Diversity: A Differential Dual-Path MoE Solution for Multilingual Neural Machine Translation

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ABSTRACT

Multilingual Neural Machine Translation (MNMT) aims to develop a unified model that enables knowledge sharing across languages to improve translation quality. In the field of MNMT, multilingual interactions exhibit two aspects: cross-lingual knowledge sharing and inter-language interference, while existing approaches always regard these two aspects as opposite, hindering overall performance. To reconcile the above two aspects, we propose a Commonality-Diversity Mixture of Experts (CDMoE) model to mitigate interference while enhancing knowledge sharing effectively. Within this method, the cross-lingual perception pathway (CLPP) captures cross-lingual shared features to enhance knowledge transfer, while the independent specialized expert pathway (ISEP) employs an explicitly segmented approach to group linguistically similar languages, enabling language-specific feature learning. This design effectively integrates cross-language knowledge sharing and inter-language interference mitigation, maintaining multilingual synergy. Moreover, it leads to superior translation quality, especially for low-resource languages. Our method achieves state-of-the-art performance on MNMT datasets, improving translation efficiency without increasing parameters.

1. Introduction

Multilingual Neural Machine Translation (MNMT) facilitates translation across multiple language pairs through a unified architecture. Based on the analysis of existing research, as illustrated in Figure 1, we identify two fundamental interaction mechanisms: (1) cross-lingual synergy fosters shared contextual understanding. This effect is particularly beneficial for low-resource languages, as it facilitates knowledge transfer from high-resource counterparts. This process also enables the effective acquisition of universal linguistic structures (e.g., grammatical rules and syntactic patterns) through joint training (Siddhant, Johnson, Tsai, Ari, Riesa, Bapna, Firat and Raman, 2020). (2) Conversely, the joint training of multiple languages may induce linguistic interference (Dabre, Cromieres and Kurohashi, 2017), especially between typologically similar languages (e.g., Spanish-Portuguese) (Huang, Li, Ma, Yao and Liu, 2023). This interference manifests as negative transfer, particularly evident when co-training high- and low-resource languages, where dominant high-resource languages constrain low-resource language acquisition (Firat, Sankaran, Al-onaizan, Yarman Vural and Cho, 2016). In light of this, the trade-off between cross-lingual synergy and interference seems to play a pivotal role in model performance.

Despite the advancement of existing MNMT methods, the two types of impact mentioned above seem to be always in opposition to each other (Tan, Leng, Chen, Ren, Qin and Liu, 2019). Therefore, if it is possible to break this situation and enable the model to share contextual knowledge across languages while simultaneously mitigating the interference between them, then the coordination of these two aspects will facilitate the development of a robust mechanism, thereby enhancing MNMT performance.

To achieve this goal, we propose a **C**ommonality-**D**iversity **M**ixture **o**f **E**xperts (**CDMoE**) model applied to the MNMT field(Yang, Yin, Ma, Zhang, Li and Wei, 2022). Unlike conventional MoE architectures that rely on a single unified routing pathway, our framework explicitly decomposes multilingual modeling into two functionally distinct yet complementary processing pathways: a cross-lingual perception pathway (CLPP) captures cross-lingual common features, while an independent specialized expert pathway (ISEP) handles language-specific nuances. Specifically, the CLPP facilitates the acquisition of shared contextual knowledge and enhances cross-lingual knowledge transfer. In the ISEP, in order to mitigate inter-language interference and enhance language-specific adaptation, we introduce the concept of explicitly segmented experts and routing. This separation enables each expert to specialize in distinct linguistic features, minimizing cross-lingual interference and enhancing language-specific representation. Additionally,

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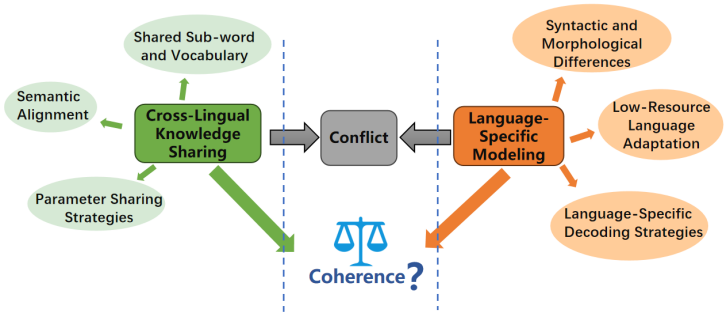


Figure 1: The figure illustrates the conflict and potential coherence between cross-lingual knowledge-sharing strategies and language-specific modeling approaches.

explicit segmentation may isolate language representations. We propose a clustering-guided explicitly segmented expert architecture that enhances expert specialization through a two-stage routing mechanism. By aggregating tokens based on linguistic correlations and selectively assigning them to experts, our approach improves domain-specific representation learning. The final output combines weighted contributions from both pathways, which advances MNMT(Zhang, Bapna, Sennrich and Firat, 2021) performance by systematically coordinating complementary learning objectives through differentiated expert processing.

We evaluate the proposed method on multiple mainstream MNMT (Lin, Wu, Wang and Li, 2021) datasets. The experimental results on IWSLT, WMT and FLORES-200 show that our method outperforms existing MNMT approaches (Tan, Wu and Monz, 2024), yielding average improvements of 2.23 BLEU4 / 3.13 chrF++ on IWSLT, 1.87 BLEU4 / 2.77 chrF++ on WMT, and 1.47 BLEU4 / 2.80 chrF++ on FLORES-200, respectively. In addition, we further report the bidirectional translation performance of our model on the KDE4 dataset. In particular, our model significantly demonstrates the effectiveness of expert specialization and cross-lingual knowledge transfer in diverse multilingual settings. The key contributions are:

1. We propose a heterogeneous dual-path MoE model integrating two pathways: a cross-lingual perception pathway for shared feature learning and an independent specialized expert pathway for capturing linguistic commonalities. This design achieves an optimal coherence between cross-lingual generalization and language-specific specialization.

2. We propose a language-aware token aggregation framework combined with competitive expert routing, explicitly preserving expert specialization across languages. By enforcing clear expert responsibility boundaries while adapting routing decisions to linguistic structures, our design ensures accurate token-to-expert alignment and avoids expert homogenization.

3. We evaluated our method on mainstream MNMT datasets. Our method achieves state-of-the-art performance, significantly improving both high- and low-resource translation without increasing model parameters, offering new insights for MoE in multi-domain learning.

2. Related Work

2.1. Research on MNMT

Compared to monolingual NMT, MNMT improves training efficiency, reduces model size, and facilitates cross-lingual knowledge transfer (Zoph, Yuret, May and Knight, 2016). MNMT builds a unified model that shares parameters across language pairs, enhancing translation quality, particularly for low-resource languages (Lakew, Federico, Negri and Turchi, 2018; Wang and Zhang, 2022). In MNMT, parameter sharing can lead to both facilitation and interference. cross-lingual complementarities aid disambiguation, enhancing translation quality (Zoph and Knight, 2016). Interlanguage facilitation is also reflected in the fact that low-resource languages can significantly improve translation performance by sharing knowledge from high-resource lexical languages (Zoph et al., 2016; Sarkar, Das, Nath and Mukhopadhyay, 2024). The performance is especially significant in language pairs with grammatical and lexical similarities (Kunchukuttan and Bhattacharyya, 2017). However, the problem of inter-language interference should not be ignored. This issue is particularly pronounced when training models with a large number of heterogeneous languages, as the dominance of high-resource languages tends to suppress the performance of low-resource languages (Johnson, Schuster, Le, Krikun, Wu, Chen, Thorat, Viégas, Wattenberg, Corrado et al., 2017). In addition, excessive

differences between different languages may lead to conflicting optimization directions of model parameters, further reducing the overall translation quality (Goyal, Gao, Chaudhary, Chen, Wenzek, Ju, Krishnan, Ranzato, Guzmán and Fan, 2022). To date, most of the current approaches focus on one aspect of enhancing multilingual knowledge sharing or reducing inter-language interference, but no better solution can take both into account (Huo, Feng, Huang, Fu, Wang and Qin, 2024; Tan et al., 2019). Therefore, how to reasonably model the shared information and differences between languages at the same time is still an important issue.

2.2. Research on MoE

MoE is increasingly at the center of LLM development (Pedicir, Miller and Robinson, 2024). The approach introduces multiple expert sub-models and dynamically allocates input data to these experts through routing. MoE can efficiently allocate model capacity and reduce unnecessary computation,(Pires, Schmidt, Liao and Peitz, 2023) which is particularly suitable for tasks that need to process large-scale and diverse data, such as multilingual translation and natural language understanding (Zoph, Bello, Kumar, Du, Huang, Dean, Shazeer and Fedus, 2022). For the above MNMT tasks, MoE can provide an efficient sparse activation scheme to dynamically select the most suitable expert module for each language through routing, so that the model can better adapt to the characteristics of the language (Team et al., 2024). It has been theoretically demonstrated that MoE routing networks can learn clustering center features (Chen, Deng, Wu, Gu and Li, 2022), which matches the multilingual nature of the MNMT problem (Zhou, Lei, Liu, Du, Huang, Zhao, Dai, Le, Laudon et al., 2022; Do, Khiem, Pham, Nguyen, Doan, Nguyen, Liu, Ramasamy, Li and Hoi, 2023). We strategically adapt these core design principles to develop a systematic framework for addressing the persistent challenges in Multilingual Neural Machine Translation (MNMT) systems. To the best of our knowledge, we are the first to propose an MoE routing mechanism that dynamically coordinates shared and language-specific features, effectively reducing mutual interference while facilitating information sharing.

3. Method

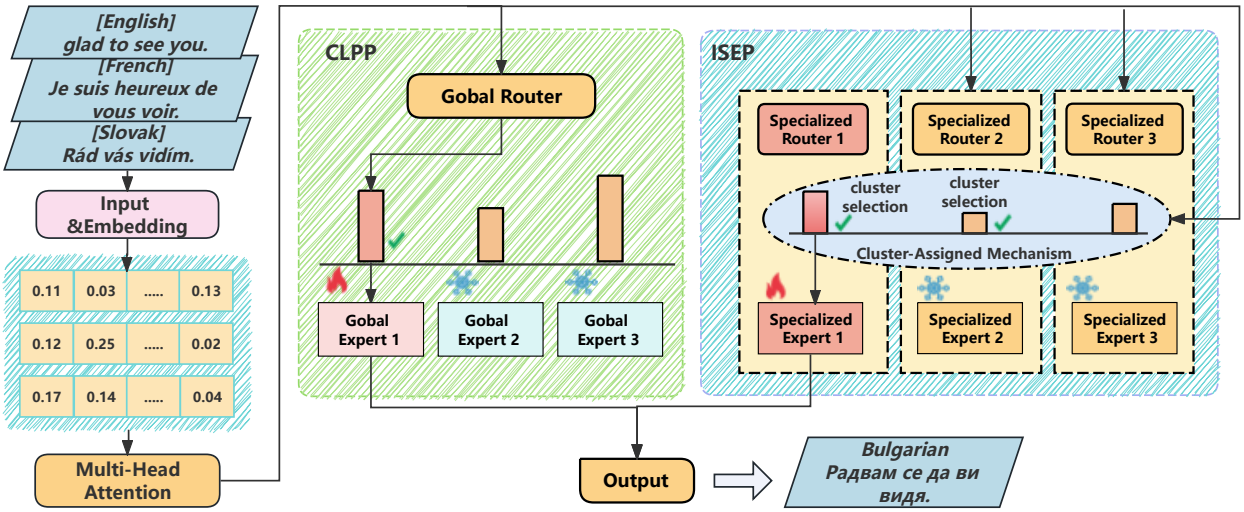


Figure 2: The Architecture of the proposed CDMoE Model. CLPP follows a classic MoE architecture, while ISEP employs explicit expert segmentation combined with clustering-guided assignment to enhance language-specific modeling. Each specialized router corresponds to a cluster center. For a given input, the nearest n clusters are identified, and the specialized routers score the input accordingly.

In this section, we elaborate on our proposed method in three stages. First, we introduce the encoding component. Next, we provide a detailed discussion of the two pathways in separate subsections. Finally, we present the fusion mechanism that integrates the output features from both pathways. As is shown in Figure 2. We innovatively design two parallel expert pathways: the ISEP and the CLPP, in order to realize the effective combination of inter-language knowledge sharing and focused learning of linguistic features. Details of each module are given respectively in the remainder of this section.

3.1. Encoding Strategy

First, we tokenize the source and target language corpora using Byte Pair Encoding (BPE) (Bostrom and Durrett, 2020), which segments the text into subwords and optionally converts it into token IDs. BPE facilitates partially shared vocabularies across languages and improves the model's performance on low-resource languages. The resulting token sequences are then mapped to integer IDs based on the vocabulary and used as input to the neural network model. We encode the input using a multilingual pre-trained Transformer model, extracting the final-layer hidden states as semantic representations. We employ a standard MLM-based multilingual encoder trained on the top 104 Wikipedia languages (Lee and Toutanova, 2018). During encoding, we add language-specific special tokens (e.g., <lang>) to help the model distinguish different target languages.

3.2. Cross-lingual Perception Pathway

In this section, we introduce the cross-lingual perception pathway. In the MoE model, the decision of which experts each token should be assigned to is usually made through a routing network (i.e., Router). The Router calculates the score assigned to each expert for each input token:

$$g_i(x) = \text{softmax}(W_g x + b_g) \tag{1}$$

Where the input feature is x, W_g is the weight matrix of the router, and b_g is the bias term. $gi(x)$ is the probability that the input token is assigned to *expert i*. The input token is assigned to *expert i* only. Each token is assigned to only the top-k scoring experts and is normalized. The formula is as follows:

$$\begin{aligned} &\text{Top-}k(x) = \{i_1, i_2, \ldots, i_k\}, \\ &\qquad \text{where } g_{i_1}(x) \geq \cdots \geq g_{i_k}(x) \geq g_{i_m, m \notin Top-k}(x) \end{aligned} \tag{2}$$

$$\hat{g}_i(x) = \begin{cases} \dfrac{g_i(x)}{\sum_{j \in Top\text{-}k(x)} g_j(x)}, & \text{if } i \in Top\text{-}k(x) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

Each expert processes the input independently to get the output:

$$y_i = f_i(x) = \sigma(W_i x + b_i), \quad y = \sum_{i \in \text{Top-}k(x)} \hat{g}_i(x) \cdot f_i(x) \tag{4}$$

Where $f_i(x)$ is the forward computation logic (e.g., linear transformation plus activation function) for *expert i*, the final output is a weighted sum of the outputs of all selected experts. The CLPP serves as a globally common knowledge modeling module. It efficiently captures common features and semantic representations across languages using shared parameters, providing stable translation support for all languages.

3.3. Independent Specialized Expert Pathway

The independent specialized expert pathway (ISEP) consists of two key components: an expert competition network for selective assignment and a language-aware distribution module for token aggregation.

Expert Competition Router We implement a competitive routing strategy where specialized routers compute relevance scores for each token. The token is then routed to the specialized experts linked to the highest-scoring specialized router. Each specialized expert has a dedicated router, implemented as an independent network. This scoring mechanism mitigates clustering inaccuracies: even if a token is assigned to multiple candidate clusters (e.g., top-3 nearest centers), the competitive scoring ensures optimal expert allocation. (Huang, An, Zhuang, Tao, Zhang, Jin, Xu, Chen, Huang and Feng, 2024)

Language-Aware Distribution Module To enhance expert differentiation, we employ a clustering-guided strategy, facilitating more precise token allocation to the most suitable specialized expert pathway. Before clustering, we perform dimensionality reduction on the embeddings using PCA. The clustering assignments are established prior to training, with PCA projections and cluster centers computed once to provide stable and consistent expert responsibilities

throughout optimization. The clustering mechanism employs a squared Euclidean distance between data points and cluster centers, where each center corresponds to a distinct specialized router and its associated specialized expert. For each input token, the system selects the top-n nearest cluster centers. Clustering is conducted at the token level, where each encoder token embedding serves as an individual data point for clustering and expert routing.

$$\min_{\{\mu_i\}_{i=1}^{K}} \sum_{i=1}^{K} \sum_{x \in S_i} ||x - \mu_i||^2 \tag{5}$$

As shown in equation 6, where K is the number of clusters, S_i denotes the ith cluster, μ_i is the center (center of mass) of the cluster S_i, The clustering centers are given by the MLM pre-training model described above, and each clustering center corresponds to a different language cluster. x is the data point belonging to the cluster S_i, and $||x-\mu_i||^2$ denotes the square of the Euclidean distance of the data point to the center of the cluster.

This design achieves two critical objectives: (1) Language-specific specialization: The ISEP focuses on linguistic patterns within their assigned clusters, enhancing adaptation to language-specific features; (2) Interference reduction: Physical separation of cluster-specific processing paths minimizes cross-lingual interference. The combined effect significantly improves multilingual modeling capacity while maintaining efficient parameter utilization.

$$A_{ij} = \begin{cases} 1, & \text{if } i = C_j \text{ and } i = \arg\max_{j'} S_{ij'} \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

The mechanism of ISEP can be described by Equation (6). $C \in \{1, 2, \ldots, j, \ldots, m\}^m$, C_j denotes the clustering result of the jth token (i.e., which category the jth token belongs to). S_{ij} denotes the scoring of the jth token by the ith specialized router network. Since the clustering centers correspond one-to-one with the expert routes, this also represents that the jth token is assigned to the ith clustering cluster. A_{ij} indicates whether the jth token is assigned to the ith network. A_{ij}=1 means assigned and A_{ij}=0 means not assigned.

3.4. Pathway Fusion

In our MNMT framework, the CLPP and the ISEP process input sequences independently before undergoing weighted fusion:

$$H_{\text{fusion}} = \alpha \cdot H_{\text{sub}} + (1 - \alpha) \cdot H_{\text{main}} \tag{7}$$

Where α is the learnable weight parameter in the range [0,1]. The language-specific expert output separates feature representations denoted as H_{sub} and H_{main} output of the CLPP. Despite sharing identical dimensionality, these representations encode complementary linguistic aspects.

4. Experimental Setup

4.1. Datasets and Assessment Indicators

To validate the effectiveness of the CDMoE model, we conducted experiments on commonly used datasets in MNMT. These experiments include both high- and low-resource language pairs to ensure coverage of a wide range of languages and resource levels.

Datasets In our experiments, we utilize widely adopted multilingual translation benchmarks to evaluate the performance of our models. Specifically, we employ the IWSLT17 translation task with augmented training data and mixed-year WMT translation benchmarks for En2De, En2Fr, and En2Ro, together with FLORES-200 for English-to-multilingual translation. For WMT, different benchmark years are used for different language pairs according to data availability. Since many of the corpora listed in Tables 1 and 2 (e.g., OpenSubtitles, ParaCrawl, CCMatrix) are drawn from public parallel data collections rather than official core releases, we treat them as extended training resources for the corresponding translation tasks, following common practice in prior MNMT studies. This enables us to construct comparable training sets for En2De, En2Fr, and En2Ro while covering multiple domains such as spoken language, news text, parliamentary proceedings, and subtitles. We use IWSLT17 instead of IWSLT24 because the latter does

Corpus	En2De	En2Fr	En2Ro
TED Talks / WIT3 (Cettolo, Girardi and Federico, 2012)	206K	230K	220K
MultiUN (Eisele and Chen, 2010)	-	13.2M	-
OpenSubtitles (Lison and Tiedemann, 2016)	12.0M	20.0M	1.1M
NewsCommentary (Tiedemann, 2012)	300K	330K	30K
Tatoeba (Tiedemann, 2012)	200K	300K	200K
Total	**12.9M**	**34.4M**	**1.55M**

Table 1
Training data statistics for the IWSLT17-based multilingual setting with extended parallel corpora across En2De, En2Fr, and En2Ro.

Corpus	En2De	En2Fr	En2Ro
Europarl (Koehn, 2005)	1.9M	2.0M	0.4M
NewsCommentary (Tiedemann, 2012)	0.3M	0.33M	0.03M
ParaCrawl (Bañón, Chen, Haddow, Heafield, Hoang, Esplà-Gomis, Forcada, Kamran, Kirefu, Koehn et al., 2020) / CommonCrawl / CCMatrix	8.0M	12.0M	0.6M
OpenSubtitles (Lison and Tiedemann, 2016)	12.0M	20.0M	1.1M
Giga / News Crawl	6.3M	4.9M	0.8M
SETIMES2	-	-	0.5M
Tatoeba / Wiki Titles / Misc. (Tiedemann, 2012)	0.2M	0.3M	0.2M
Total	**29.7M**	**39.8M**	**3.6M**

Table 2
Mixed-year WMT statistics (En2De from WMT24, En2Fr from WMT14, En2Ro from WMT16).

not provide complete parallel data for En2De, En2Fr, and En2Ro, while IWSLT17 includes all three language pairs. To better evaluate performance in low-resource and zero-shot translation settings, we additionally employ the KDE4 dataset. KDE4 contains high-quality parallel data from software localization files and is commonly used to assess multilingual translation under limited-data conditions. This dataset combination supports evaluation across different resource levels and translation scenarios.

Assessment Indicators Our method is evaluated using the BLEU4 metric. BLEU4 is the most widely used automatic metric. It evaluates translation quality by measuring the similarity between generated text and reference translations. Specifically, BLEU4 calculates a weighted geometric mean of four n-gram matches to assess fluency and accuracy (Papineni, Roukos, Ward and Zhu, 2002). The BLEU4 score usually ranges from 0 to 100, and the higher the score is, the closer the quality of the translation is to the reference text.

4.2. Baseline Model

In our experiments, our method is compared with the current optimal or most commonly used MNMT methods, including m-Transformer (Johnson et al., 2017), StableMoE, mBART, and mRASP, to validate its effectiveness in multilingual translation tasks, especially in low-resource language Improvement.

StableMoE is an optimized Mixture of Experts (MoE) variant designed to improve training stability, expert load balancing, and computational efficiency (Dai, Dong, Ma, Zheng, Sui, Chang and Wei, 2022).

Multilingual BART (mBART) is a multilingual autoregressive text denoising pre-training model proposed by Facebook AI (Chipman, George, McCulloch and Shively, 2022), based on BART (Tang, Tran, Li, Chen, Goyal, Chaudhary, Gu and Fan, 2020). It is pre-trained on a large-scale multilingual unsupervised corpus with a denoising autoencoder, which can provide strong cross-language migration capability in MNMT tasks, especially for translating low-resource languages.

mRASP is an improved multilingual pretraining method that enhances cross-lingual transfer by aligning representations across languages (Pan, Wang, Wu and Li, 2021). It leverages translation-based denoising and token reordering to improve both high- and low-resource translation performance.

M2M-100 is a fully multilingual many-to-many translation model developed by Meta AI, supporting direct translation between 100 languages without relying on English as an intermediate pivot (Fan, Bhosale, Schwenk, Ma, El-Kishky, Goyal, Baines, Celebi, Wenzek, Chaudhary et al., 2021). It is trained on the large-scale M2M parallel corpus with a balanced multilingual sampling strategy.

NLLB-200 (No-Language-Left-Behind) is another large-scale multilingual MT model introduced by Meta AI (**?**), covering over 200 languages with improved quality for diverse language families. It employs a dense transformer architecture with advanced data filtering, language identification, and vocabulary balancing techniques. To ensure fair comparison in our experiments, we adopted a consistent approach for all baselines. We fine-tuned each baseline model on the same training data and corresponding splits that were used for our proposed model. However, this fine-tuning was performed only when the baseline had not already been pre-trained on those specific datasets. If a baseline had already been pre-trained on the relevant data, we did not apply further fine-tuning. This was done to avoid redundant optimization and to preserve the model's original multilingual capabilities. For instance, large-scale models like M2M-100 and NLLB-200, which were pre-trained on extensive multilingual corpora including the languages and domains covered in our benchmarks.

4.3. Environment and Parameter Setting

We adopt the Adam optimizer (Reddi et al., 2019) with betas set to (0.9, 0.98) and a learning rate of 0.0007. The batch is divided according to the number of tokens, with a maximum of 3,584 tokens per batch. The model is trained for up to 10 epochs or 100,000 updates, whichever comes first. Following Vaswani et al. (2017), the learning rate warms up linearly during the first 4,000 steps and then decays proportionally to the inverse square root of the update step. Our encoder and decoder are both configured using a 20-layer Transformer architecture. All embeddings are shared across the model. The encoder uses an embedding dimension of 1280, a feed-forward network (FFN) dimension of 5120, and 20 attention heads. The decoder mirrors these settings, with its embedding and FFN dimensions defaulting to those of the encoder and also using 20 attention heads. Both the encoder and decoder adopt pre-norm residual connections, with GELU selected as the activation function. For regularization, we apply a dropout rate of 0.15, an attention dropout of 0.1, and an activation dropout of 0.1. The training objective is label smoothing cross entropy, which prevents the model from becoming overconfident and improves target-language representation. In the expert architecture, the Independent Specialized Expert Pathway (ISEP) contains 3 experts, and the Cross-Lingual Perception Pathway (CLPP) also includes 3 experts. Only one expert is activated per pathway during routing.

A standard pre-processing procedure is used for all datasets. First, the raw text is subjected to word-splitting. For data splits, we strictly follow the official training, validation, and testing partitions provided by each benchmark (e.g., the standard FLORES and WMT test sets) to ensure fair and directly comparable evaluation with prior MNMT work. For datasets without predefined splits, we apply an 80%–10%–10% division to form training, validation, and testing sets, ensuring both reproducibility and consistency across experiments.

All experiments are performed on 8 NVIDIA H800 GPUs with a memory size of 80 GB, implemented using the PyTorch framework, python version 3.9, and trained using mixed precision to accelerate the experiments. For processing the language set we refer to fairseq(Ott, 2019), a framework commonly used for MNMT tasks.

5. Experimental Results

5.1. Main Experiment

Across the three evaluation benchmarks in Table 3, CDMoE exhibits consistent improvements over the strongest baseline (NLLB-200 1.3B). CDMoE yields +1.47 BLEU4 and +2.80 chrF++ improvement on FLORES-200, achieves a further +2.23 BLEU4 and +3.13 chrF++ gain on IWSLT, and delivers an additional +1.87 BLEU4 together with +2.77 chrF++ improvement on WMT. These results indicate that the proposed commonality–diversity expert design effectively enhances both token-level accuracy (BLEU4) and character-level consistency (chrF++), leading to stronger multilingual translation performance across datasets.

To further evaluate the effectiveness of our approach on multilingual translation beyond English-centric settings, we conduct experiments on the KDE4 dataset, which includes diverse language pairs across high-, medium-, and low-resource conditions. As shown in Table 4, the BLEU4 scores span a wide range, from 1.65 to 37.12, reflecting significant variation in translation quality depending on the language pair. The highest score is observed on the FR→EN direction (37.12), while the lowest occurs on AS→BN (1.65), indicating a strong correlation between data availability and translation performance. Generally, translation between high-resource languages (e.g., EN↔FR/DE) achieves higher BLEU4 scores, whereas low-resource pairs involving underrepresented languages such as Assamese (AS) and Gujarati (GU) tend to result in lower performance. These results highlight the importance of both language-relatedness and training data coverage in achieving effective multilingual translation, particularly for low-resource and non-English-centric directions.

Table 3
Model performance across FLORES-200, IWSLT and WMT datasets with BLEU4 and chrF++ metrics. Each language is shown once (merged rows). Best BLEU4 per language per dataset is **bolded**.

Language	Model	FLORES-200		IWSLT		WMT	
		BLEU4	chrF++	BLEU4	chrF++	BLEU4	chrF++
Fr	m-Transformer(150M)	30.1	52.2	30.1	53.1	30.3	48.7
	StableMoE(100M)	32.2	53.3	31.2	52.8	30.7	50.1
	mRASP(213M)	33.3	56.8	31.6	53.8	31.6	52.2
	mBART(610M)	36.7	57.9	32.8	54.2	37.1	56.0
	M2M-100(418M)	38.9	58.3	32.6	53.5	39.7	59.2
	M2M-100(1.2B)	45.1	67.5	38.7	63.6	43.1	63.8
	NLLB-200(600M)	44.3	67.0	38.3	58.1	40.5	60.2
	NLLB-200(1.3B)	45.7	68.8	43.0	65.6	44.2	65.7
	CDMoE(100M)	33.1	55.9	33.9	55.1	35.3	52.8
	CDMoE(0.9B)	**46.3**	**69.7**	**45.6**	**68.8**	**45.9**	**67.8**
De	m-Transformer(150M)	28.3	42.1	26.7	48.1	26.5	47.7
	StableMoE(100M)	27.5	46.4	24.1	44.3	27.0	48.1
	mRASP(213M)	28.2	47.2	28.7	50.1	24.7	44.4
	mBART(610M)	30.9	57.1	28.6	50.6	27.1	48.6
	M2M-100(418M)	31.2	58.4	29.1	51.7	27.3	53.2
	M2M-100(1.2B)	36.5	60.3	30.9	57.1	32.1	57.8
	NLLB-200(600M)	34.1	57.4	30.4	55.9	30.5	56.3
	NLLB-200(1.3B)	37.1	61.4	31.6	57.8	33.0	59.2
	CDMoE(100M)	27.9	46.6	29.5	46.1	27.8	48.5
	CDMoE(0.9B)	**38.7**	**64.2**	**33.2**	**59.4**	**34.5**	**62.0**
Ro	m-Transformer(150M)	24.8	41.3	23.5	46.2	24.1	47.5
	StableMoE(100M)	21.3	40.6	24.3	47.3	25.0	48.7
	mRASP(213M)	25.2	43.8	25.4	48.2	26.1	49.4
	mBART(610M)	28.9	52.4	26.7	49.9	27.6	51.1
	M2M-100(418M)	27.6	54.1	26.9	50.2	28.0	51.7
	M2M-100(1.2B)	33.8	56.9	28.6	54.1	30.3	53.9
	NLLB-200(600M)	34.2	58.2	28.9	52.8	31.0	55.0
	NLLB-200(1.3B)	35.6	59.6	30.1	54.0	32.4	56.2
	CDMoE(100M)	25.9	44.1	26.3	48.1	27.1	49.7
	CDMoE(0.9B)	**37.8**	**64.3**	**32.6**	**58.6**	**34.8**	**59.6**

5.2. Ablation Experiment

To evaluate the effectiveness of each component in our proposed model, we conduct ablation experiments on the same datasets(IWSLT) used in the main experiments, removing the global experts, specialized experts, and clustering mechanisms separately.

5.2.1. Cross-lingual Perception and Independent Specialized Expert Pathway

To evaluate the contribution of each pathway, we conduct ablation experiments by removing CLPP and ISEP separately. Removing CLPP leads to a 4.58 BLEU4 drop (29.78 → 25.20), demonstrating its essential role in facilitating cross-lingual knowledge sharing. Eliminating ISEP causes a 5.68 BLEU4 decrease (29.78 → 24.10), highlighting its importance in capturing language-specific representations. These results confirm that both pathways are essential for balancing shared knowledge modeling and language-specific adaptation. Furthermore, the combination of CLPP and ISEP yields complementary benefits, as their joint optimization achieves a better trade-off between cross-lingual transfer and specialization. This underscores the necessity of integrating both pathways to maximize overall translation performance.

Table 4
BLEU4 / chrF++ results on the KDE4 dataset across 7×7 language pairs.

	EN	DE	FR	PL	AS	BN	GU
EN	–	29.51/49.5	33.90/52.9	12.08/30.1	1.86/13.9	22.54/37.5	21.21/33.2
DE	33.37/53.4	–	23.41/43.4	7.43/25.4	2.01/14.0	20.42/35.4	3.68/15.7
FR	35.12/57.1	22.12/42.1	–	10.49/28.5	2.12/14.1	15.13/30.1	24.34/36.3
PL	16.37/36.4	8.64/28.6	13.44/33.4	–	1.97/14.0	19.72/34.7	12.11/24.1
AS	3.01/23.0	2.55/22.6	2.64/22.6	2.32/20.3	–	1.65/16.7	1.77/13.8
BN	28.61/48.6	24.03/44.0	16.23/36.2	23.80/41.8	2.11/14.1	–	10.61/22.6
GU	24.61/44.6	5.11/25.1	25.23/45.2	14.32/32.3	2.21/14.2	8.34/23.3	–

Model Variation	EN-DE	EN-FR	EN-RO	Avg
Full Model(3 experts)	**29.51**	**33.90**	**26.31**	**29.78**
Full Model(8 experts)	27.31	31.32	23.42	27.35
w/o CLPP	26.87	28.46	20.26	25.20
w/o ISEP	25.66	27.02	19.62	24.10
w/o Clustering Mechanism	26.03	29.22	20.68	25.31

Table 5
Demonstrates the impact of different modules on the results.

5.2.2. Clustering Mechanism

We remove the clustering mechanism and directly assigned tokens to specialized experts, disrupting structured expert training. As shown in the last row of Table 5, this modification results in a 4.47 BLEU4 decline (29.78 → 25.31) on average. Without structured assignments, specialized experts struggle to learn coherent language-specific patterns, leading to increased interference and reduced specialization efficiency. However, when combined with the clustering mechanism, explicit segmentation further enhances expert isolation, resulting in improved specialization and reduced interference. This result highlights the importance of clustering in maintaining expert coherence and improving translation quality.

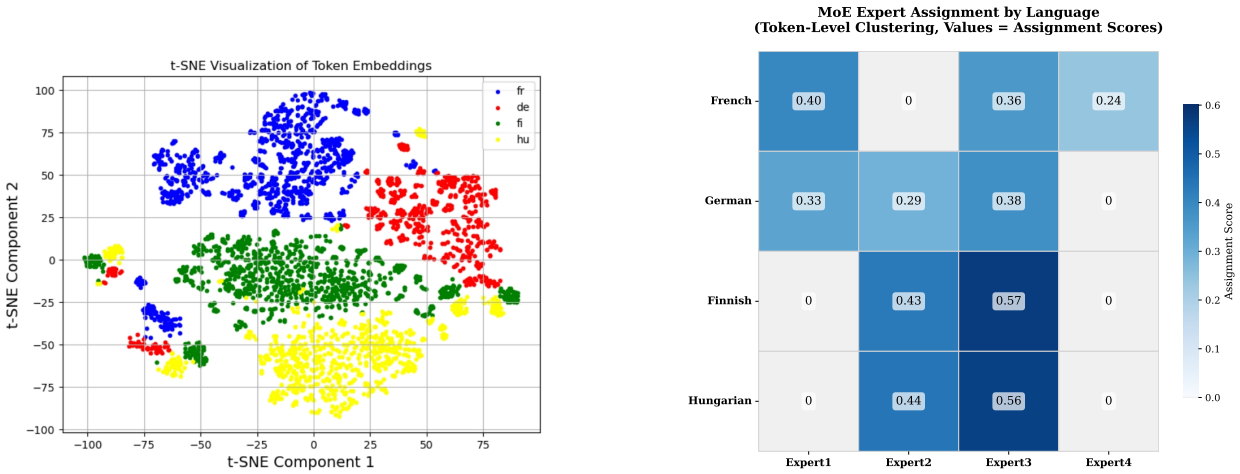
5.2.3. Experts Number

To investigate the impact of expert quantity, we compare performance between models using 3 and 8 experts. Increasing the number of experts from 3 to 8 surprisingly reduces performance by 2.43 BLEU4 (29.78 → 27.35). We consider that the regression stems from two interrelated factors: (1) With more experts, the token-to-expert assignment becomes more dispersed, leading to insufficient training for each expert and weaker specialization. (2) Expanding the expert pool increases competition among experts, amplifying routing inconsistencies and reducing translation quality. These findings highlight the importance of a well-balanced expert count in MoE-based MNMT models.

6. Analysis

6.1. Token Clustering and Distribution of Different Language Tokens

In order to demonstrate the effectiveness of the clustering methods for different languages, we select 160 sentences from each language in WMT14 dataset, with an average of 15 tokens for each language. The four different languages tested are German, French, Finnish, and Hungarian. The features are downscaled and visualized using PCA (Kurita, 2021) in combination with the t-SNE (Wattenberg, Viégas and Johnson, 2016) method, and the results of the visualization are shown in Figure 3(a). This indicates that multilingual pretraining captures language-specific structures, enabling the model to separate different languages into distinct clusters. The clear boundaries between clusters suggest that the learned representations preserve linguistic distinctions while maintaining cross-lingual relationships.



(a) Clustering of different language tokens

(b) Distribution of different language tokens among experts

Figure 3: Analysis of language token representations and expert allocation

Figure 3(b) illustrates the token-level expert assignment of our clustering-guided MoE model across four languages. The heatmap shows a clear non-uniform distribution, indicating that the routing mechanism learns meaningful expert specialization rather than relying on uniform or collapsed expert usage. Specifically, Expert 1 mainly serves French and German tokens, while Expert 2 and Expert 3 are predominantly activated by Finnish and Hungarian, with Expert 3 receiving the largest proportion of tokens from these morphologically rich languages. Such patterns are consistent with linguistic similarity, as French and German belong to the Indo-European family, whereas Finnish and Hungarian are Uralic languages with distinct morphological characteristics. In addition, several expert–language pairs exhibit near-zero activation, suggesting that the model effectively avoids assigning linguistically mismatched tokens to unsuitable experts, thereby mitigating cross-lingual interference. Overall, the observed expert selection behavior confirms that the proposed clustering-guided MoE framework successfully enables language-aware expert specialization while maintaining structured multilingual parameter sharing.

6.2. Case Study

To demonstrate the advantages of our proposed dual-pathway mechanism, which integrates both global experts and specialized experts, we present concrete translation examples illustrating how the model captures generic semantic information as well as language-specific and culture-sensitive nuances in multilingual neural machine translation.

The original sentences and their corresponding translation outputs are shown in Table 6.

Case 1. Input sentence (English): "She broke the ice with a joke, and the meeting became more relaxed."

In the NLLB-200 translation, the idiomatic expression *"break the ice"* is translated in a generalized manner, roughly conveying the meaning: "She eased the situation with a joke, and the meeting became more relaxed." While the overall meaning remains understandable, this translation abstracts away the idiomatic and pragmatic function of *"break the ice"*, which specifically refers to overcoming initial tension or awkwardness in social interactions. In contrast, the translation produced by our proposed CDMoE model accurately preserves the idiomatic expression *"das Eis brechen"*, maintaining both the semantic and pragmatic intent of the original sentence. The output is fluent, precise, and aligns well with German idiomatic usage.

Case 2. Input sentence (English): "He finally showed up, and we can start the discussion now."

In the NLLB-200 translation, the phrasal verb *"showed up"* is translated in a literal or overly formal manner, roughly conveying the meaning: "He finally appeared, and we can start the discussion now." Although this translation is grammatically correct and conveys the general idea, it weakens the conversational register and the pragmatic implication of *"showed up"* in this context (i.e., arriving after being expected, often with an implicit sense of delay or relief). In contrast, the CDMoE output uses a more context-appropriate and natural German expression (e.g., *"ist endlich angekommen"*), better preserving the informal tone and the pragmatic nuance of the source sentence. This improves both fluency and faithfulness to the original communicative intent.

Sentence	Text
Case 1 Input	She broke the ice with a joke, and the meeting became more relaxed.
Case 1 NLLB-200 output	Sie lockerte die Situation mit einem Witz auf, und das Treffen wurde entspannter.
Case 1 CDMoE	Sie brach mit einem Witz das Eis, und die Besprechung wurde entspannter.
Case 2 Input	He finally showed up, and we can start the discussion now.
Case 2 NLLB-200 output	Er erschien schließlich, und wir können die Diskussion jetzt beginnen.
Case 2 CDMoE	Er ist endlich angekommen, und wir können die Diskussion jetzt beginnen.

Table 6
Comparison of translation outputs between NLLB-200 and the proposed CDMoE model in case studies.

Overall, these examples demonstrate that the CDMoE model effectively integrates general semantic understanding with language-specific and idiomatic knowledge, leading to more accurate and context-aware translations in multilingual settings.

7. Conclusion

To reconcile the opposing aspects of enhancing multilingual knowledge sharing and mitigating inter-language interference, we propose a CDMoE for MNMT. This model mitigates language interference while improving adaptation through the synergistic interplay of global and specialized experts. The CLPP captures cross-lingual abstractions, whereas the ISEP is dynamically grouped based on latent representations in language-group patterns. This architecture effectively harmonizes linguistic generalization and specificity, fostering more robust multilingual translation. Experiments on the high- and medium-resource datasets IWSLT, WMT, and FLORES-200 show an average improvement of 1.86 BLEU4 and 2.90 chrF++, demonstrating consistent gains over existing MNMT approaches. KDE4 results further validate robustness across non-English pairs, with BLEU4 scores ranging from 1.65 to 37.12. The improvements highlight effective knowledge transfer and expert specialization in multilingual settings. We believe that the synergy between "seeking commonality while preserving diversity" opens a door to the area of MNMT, establishing a promising direction for multilingual representation learning. Future work will refine our expert assignment strategy and extend its applicability to broader tasks, including domain-specific QA systems(Nie, Wei, Zhang, Wang, Gao and Yang, 2017) (e.g., medical vs. legal) and multilingual code generation (e.g., Python-JavaScript interoperability), further enhancing adaptability across diverse linguistic and task-specific contexts.

8. Limitations

The method can significantly improve the translation quality of low-resource languages. However, when data is extremely scarce, the model may still have difficulty in learning stable and robust language representations, which negatively affects the overall translation results. Additionally, our current architecture does not explicitly include hard components for load balancing across experts. In the future, integrating our current work with advanced load-balancing techniques could be a promising and effective direction for further improvement.

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